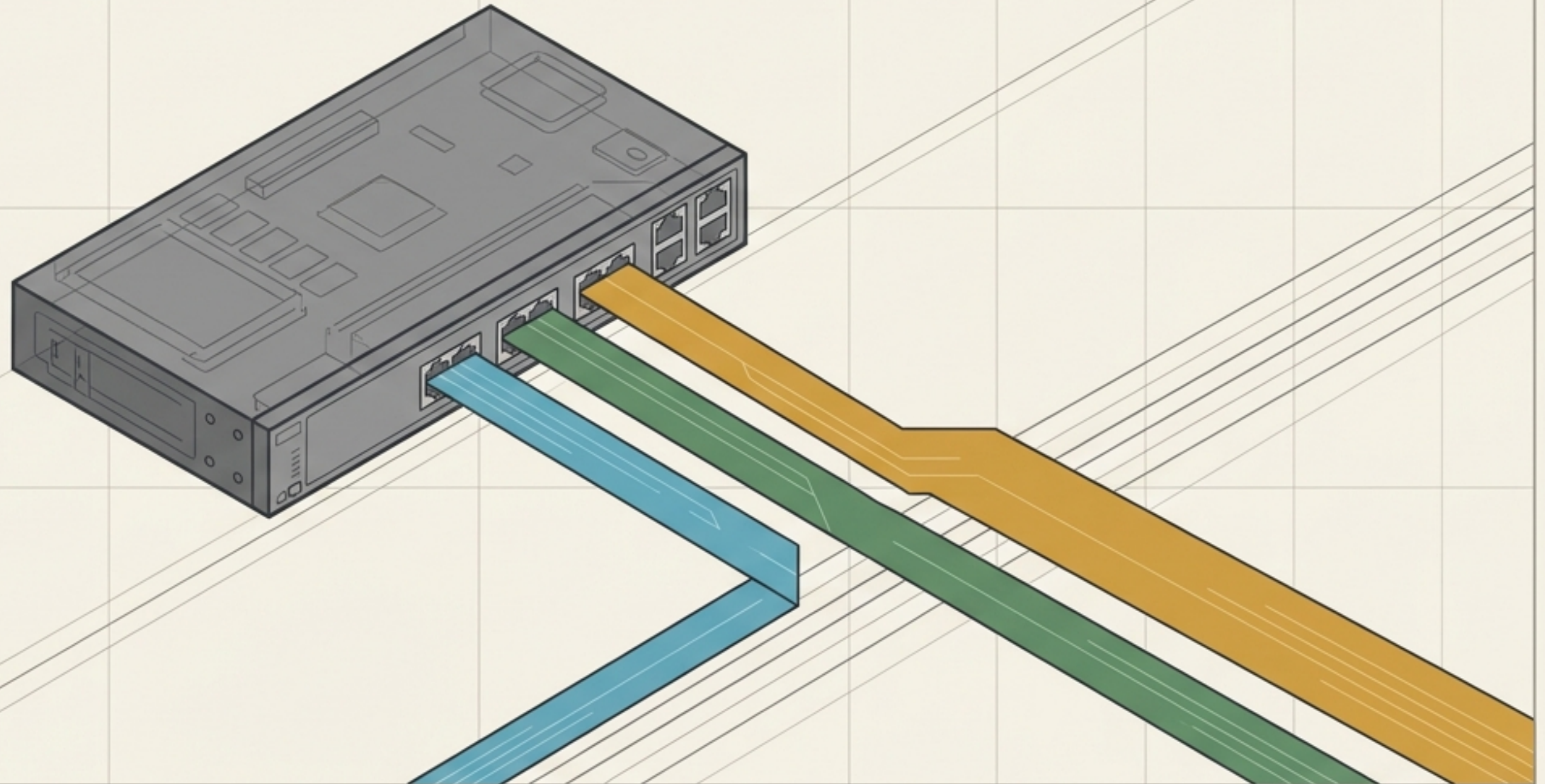


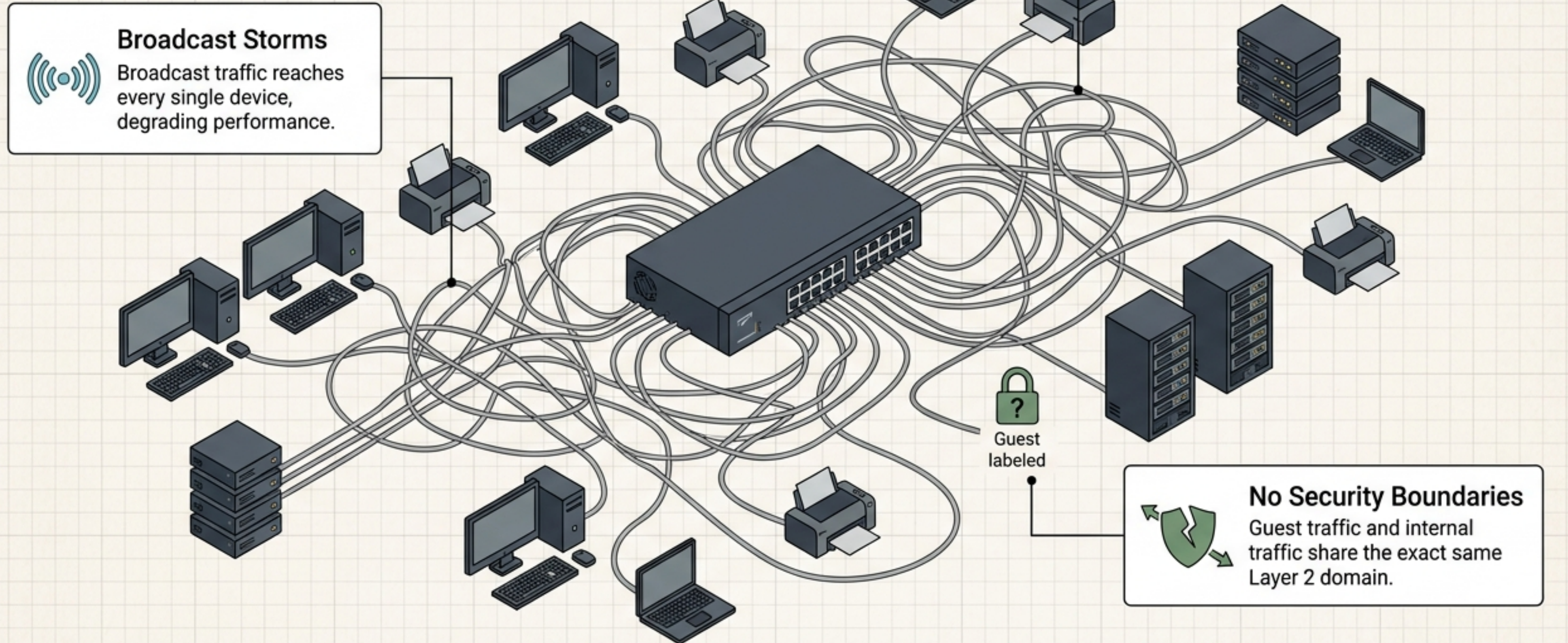
THE ARCHITECTURE OF VLANS

Fundamentals, Routing, and Security



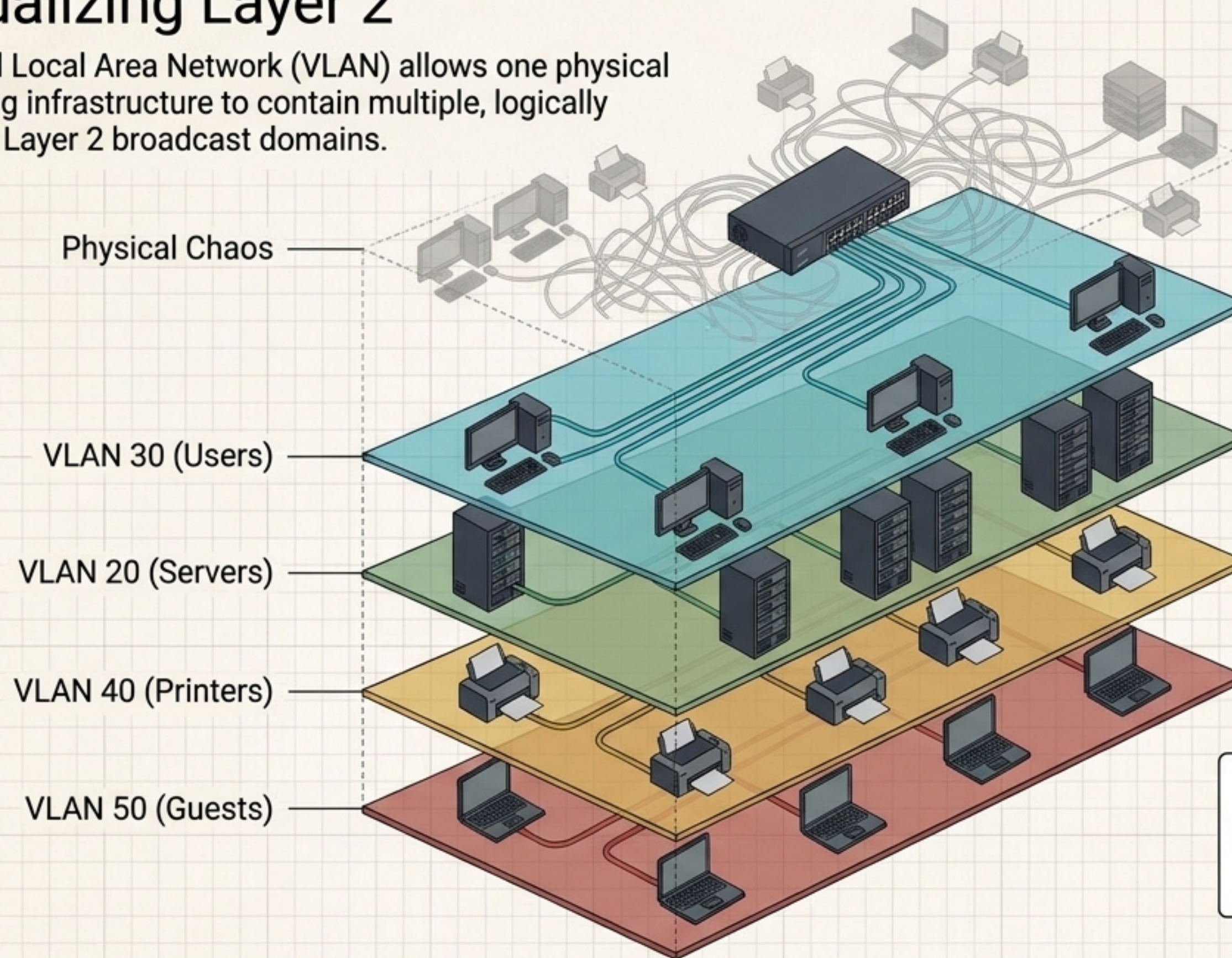
One Switch, One Domain, Total Chaos

Placing all enterprise devices into a single Layer 2 broadcast domain creates immediate operational and security hazards. The physical infrastructure is shared, but there are no logical boundaries.



Virtualizing Layer 2

A Virtual Local Area Network (VLAN) allows one physical switching infrastructure to contain multiple, logically isolated Layer 2 broadcast domains.



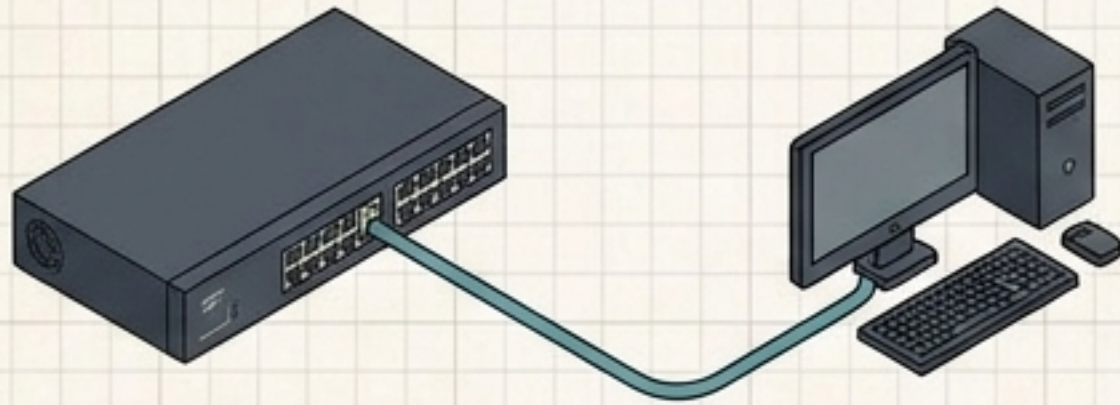
Key Insight

A broadcast generated in the User plane remains completely contained within the User plane at Layer 2.

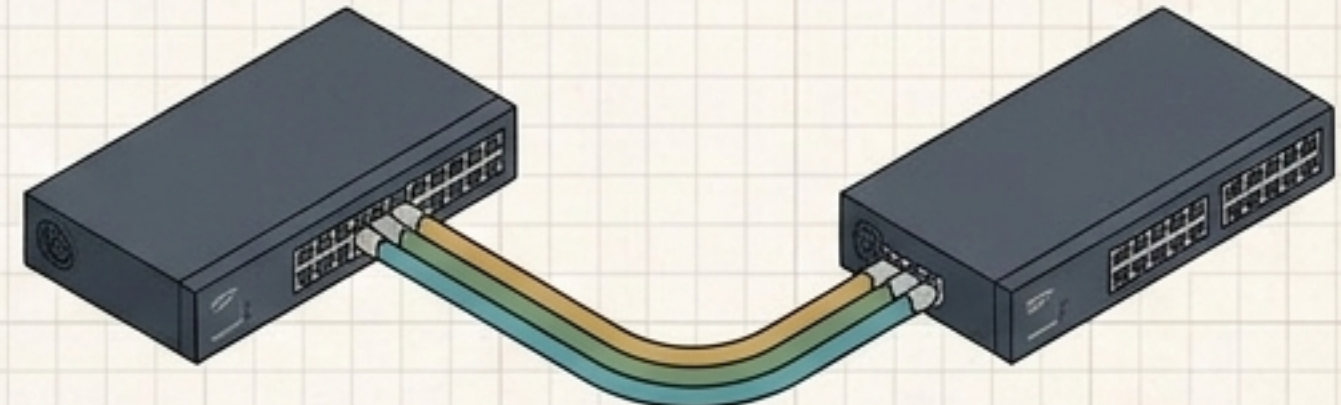
Port Designation: Access vs. Trunk

How a device connects to the switch dictates how its VLAN traffic is handled and tagged.

Access Port



Trunk Port

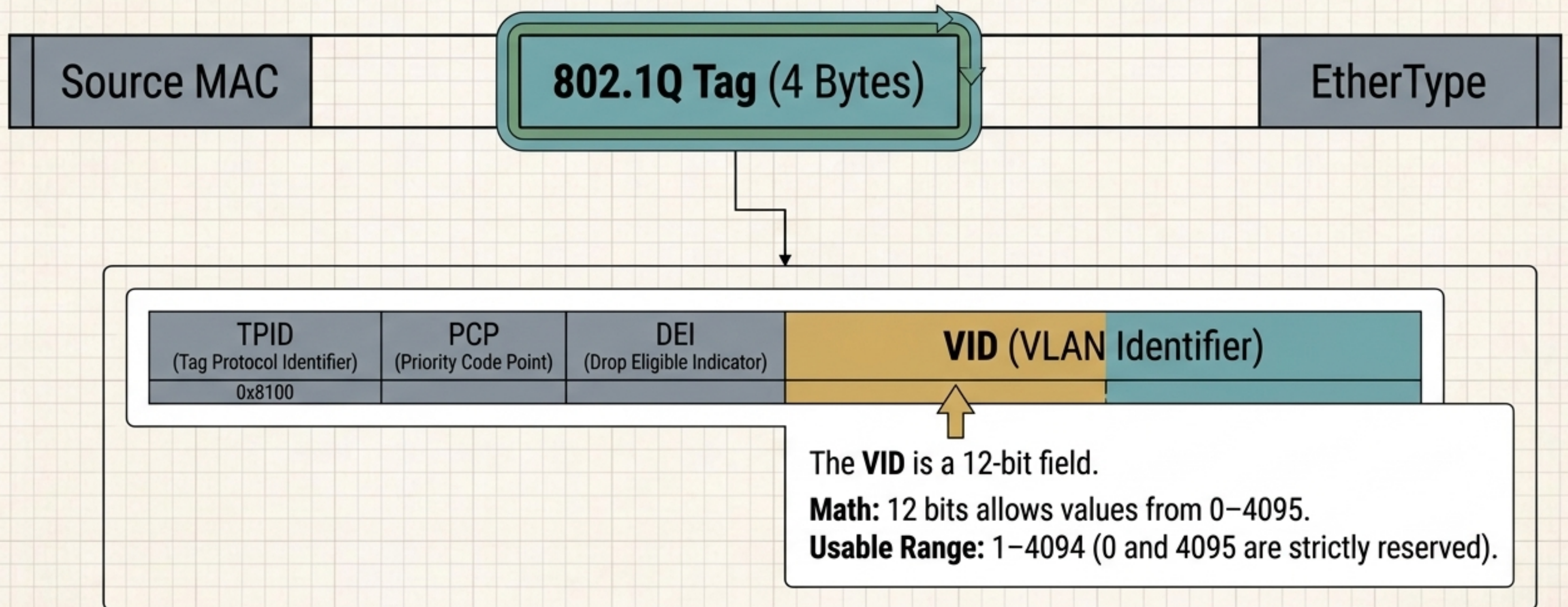


Carries:	Normally carries exactly one VLAN.
Traffic Type:	Endpoint traffic is normally untagged.
Primary Use:	Endpoint devices (PCs, Servers, Printers).

Carries:	Carries multiple VLANs over a single physical link.
Traffic Type:	Uses 802.1Q tags to identify VLAN membership.
Primary Use:	Switch-to-switch or switch-to-router links.

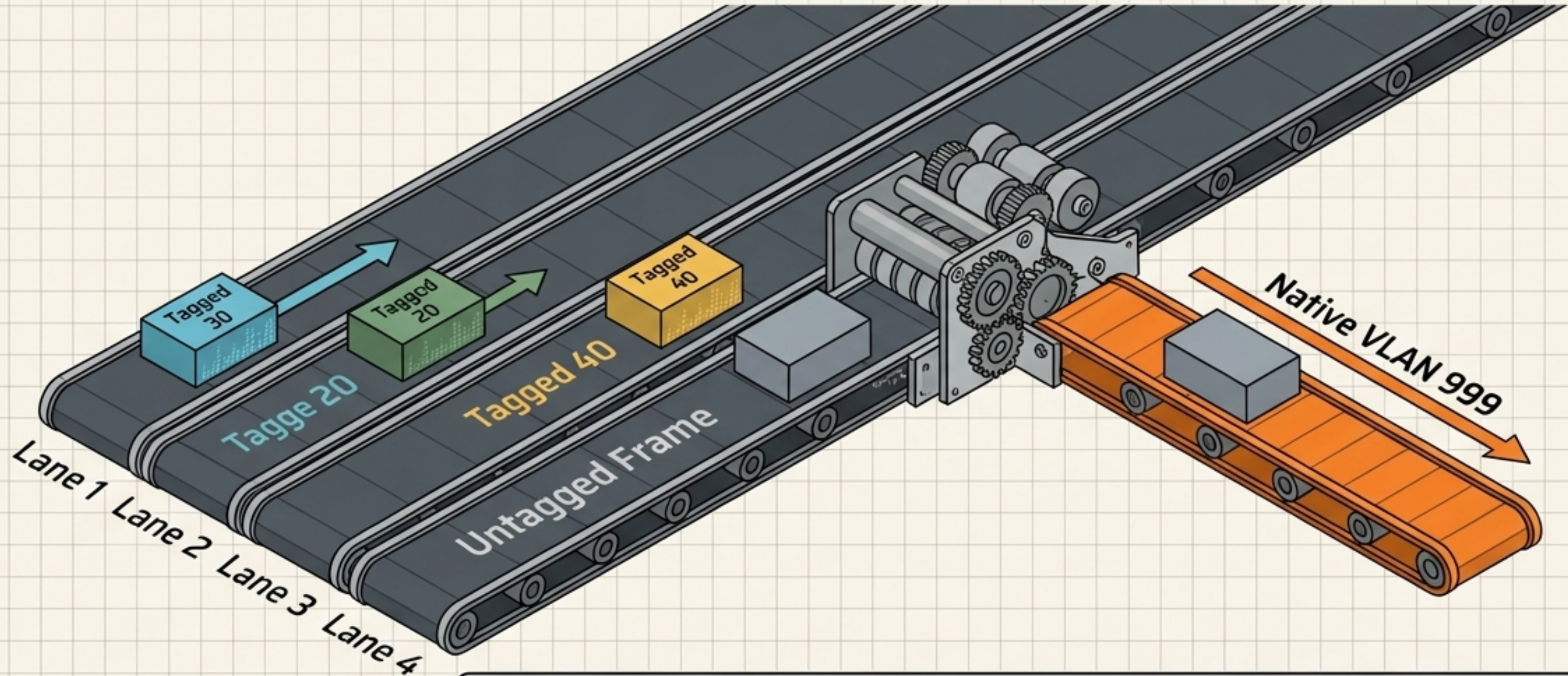
The Anatomy of 802.1Q Tagging

To multiplex multiple VLANs over a single Trunk link, switches insert a 4-byte 802.1Q tag directly into the Ethernet frame before it crosses the wire.



Handling Untagged Traffic: The Native VLAN

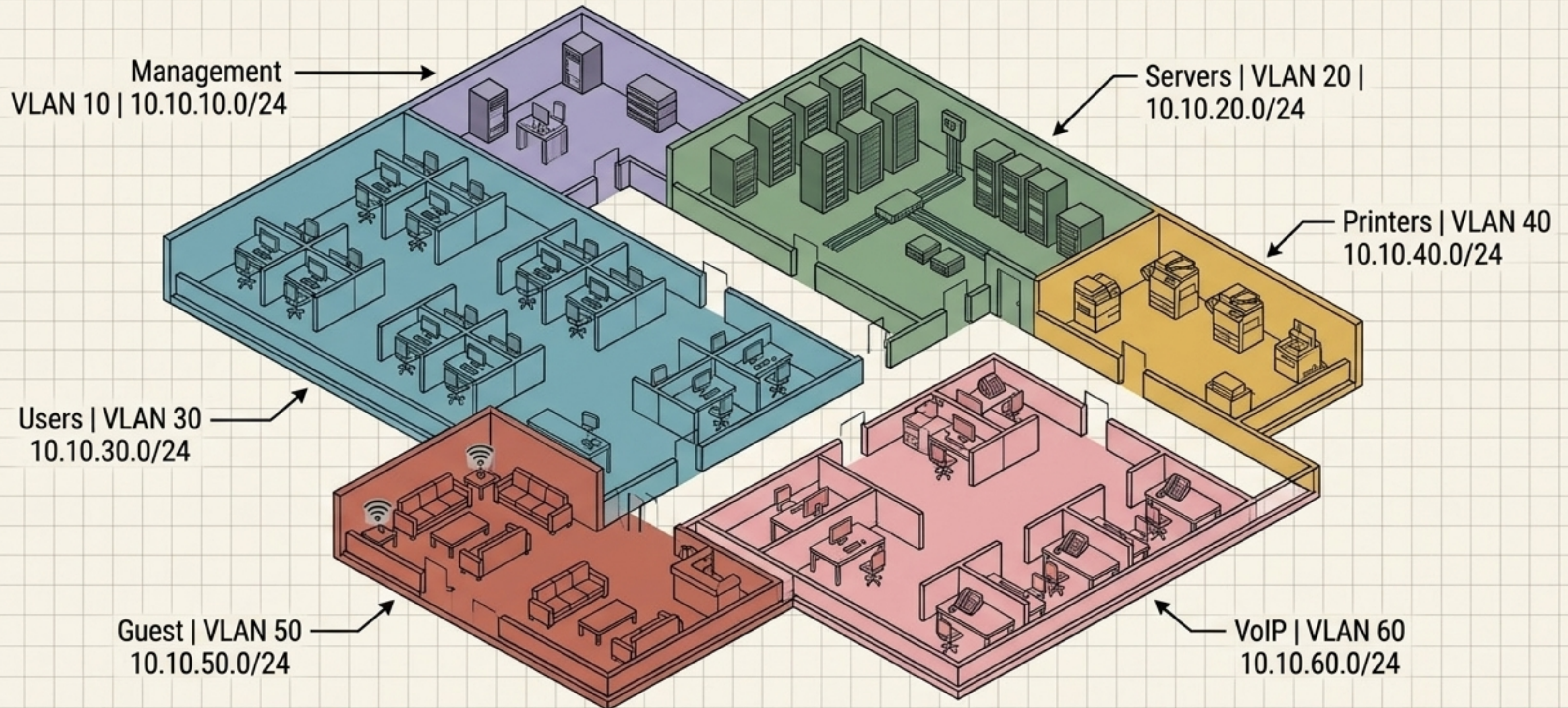
Not all frames on an 802.1Q trunk are tagged. The Native VLAN provides a defined context for untagged traffic crossing a trunk.



Warning: Both ends of a trunk must agree on the Native VLAN. A mismatch causes unexpected traffic placement and critical connectivity failures.

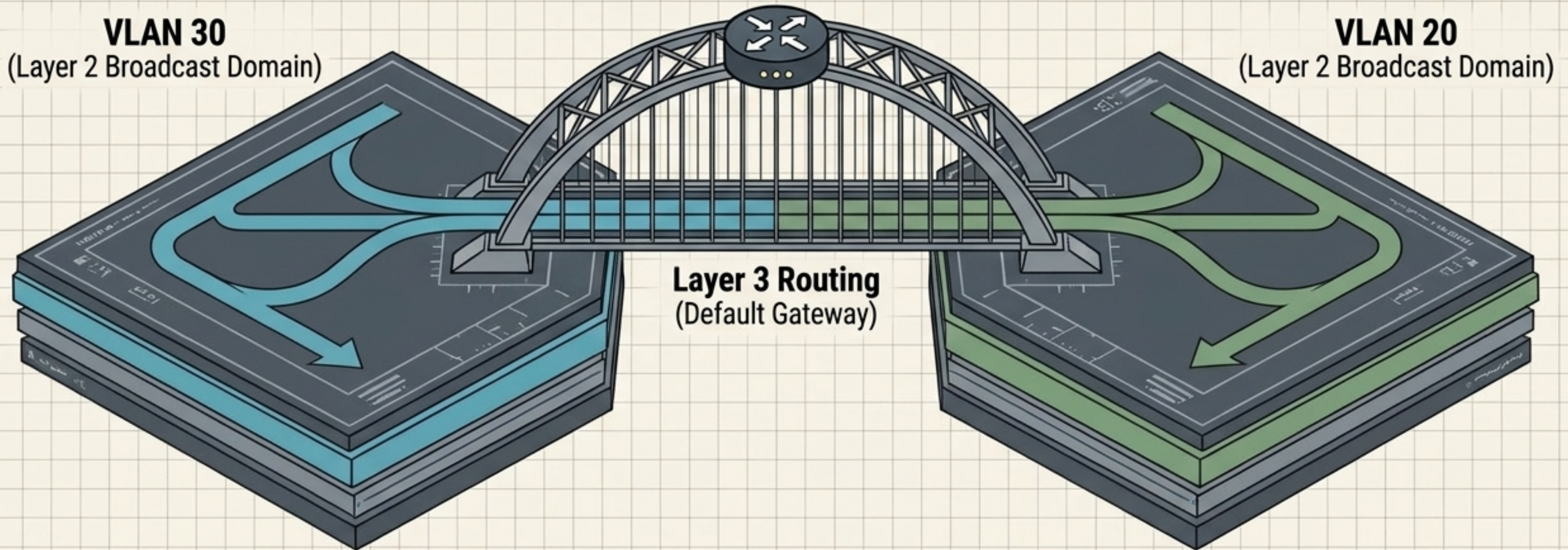
Designing the Enterprise Blueprint

Good VLAN design follows function and trust level. It establishes clean boundaries for both Layer 2 broadcast control and Layer 3 IP addressing.



The Boundary: Layer 2 vs. Layer 3

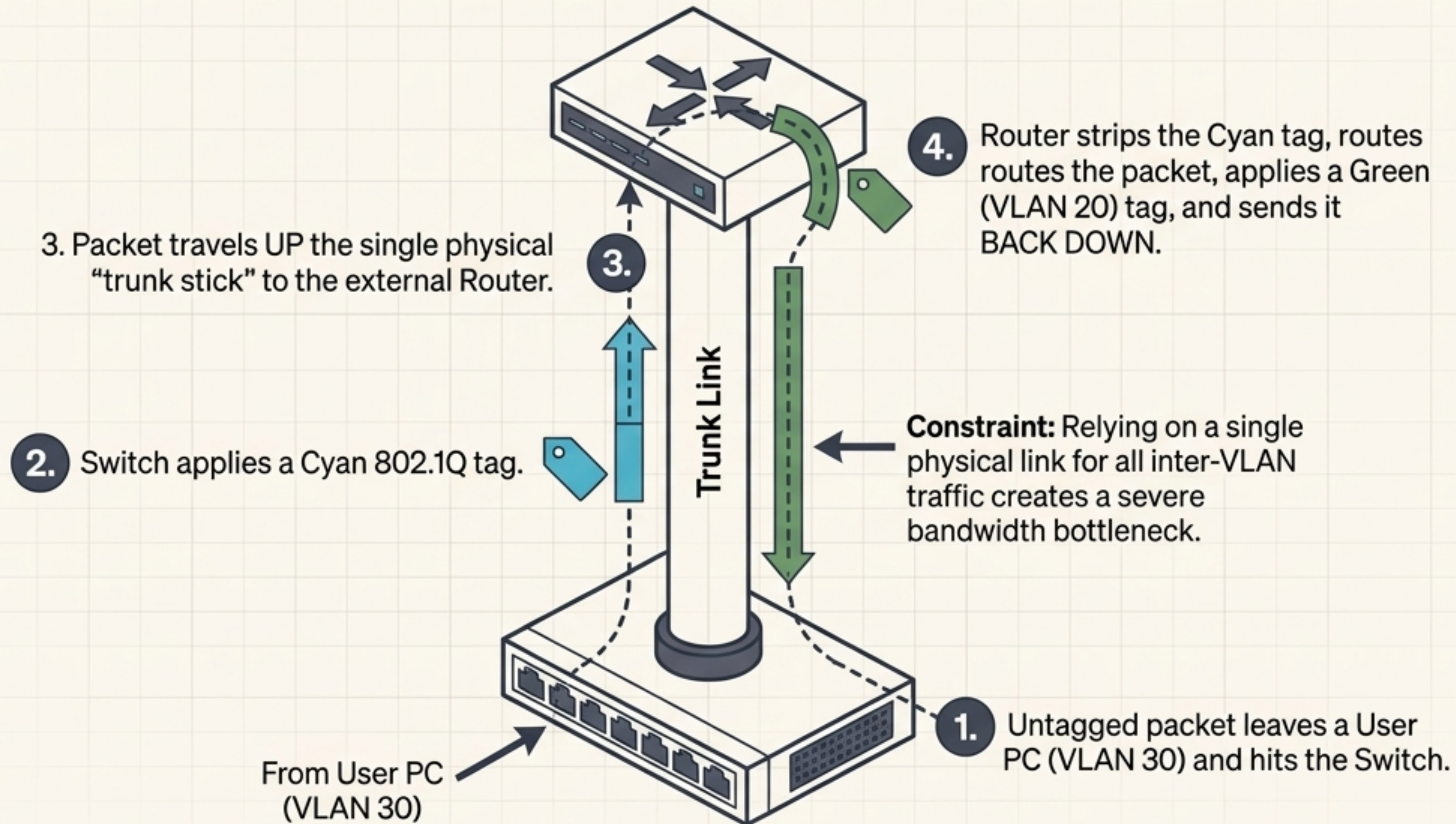
VLANs provide isolation. A device in VLAN 30 cannot send an ordinary Ethernet frame directly to a device in VLAN 20. They are logically separated.



Key Insight: VLANs do not automatically provide routing. Communication between them requires a Layer 3 gateway.

Inter-VLAN Routing: Router on a Stick

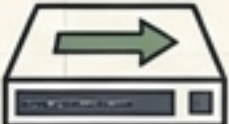
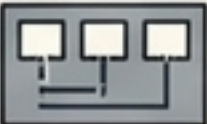
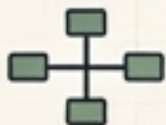
A traditional routing architecture that utilizes a single physical router interface mapped to multiple 802.1Q logical subinterfaces.

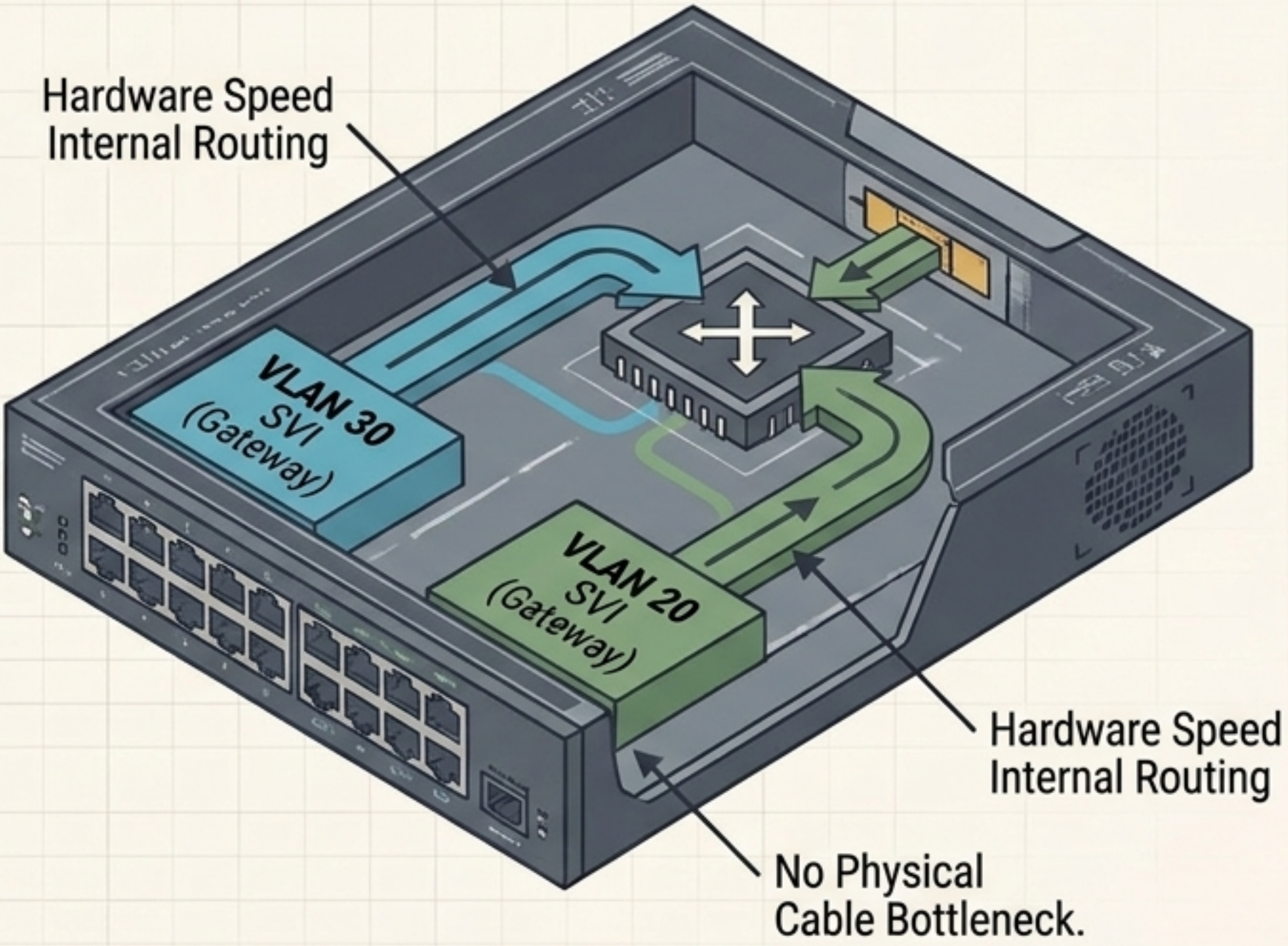


Inter-VLAN Routing: Layer 3 Switches

Modern enterprise networks perform routing internally utilizing Switched Virtual Interfaces (SVIs), eliminating the external router bottleneck.

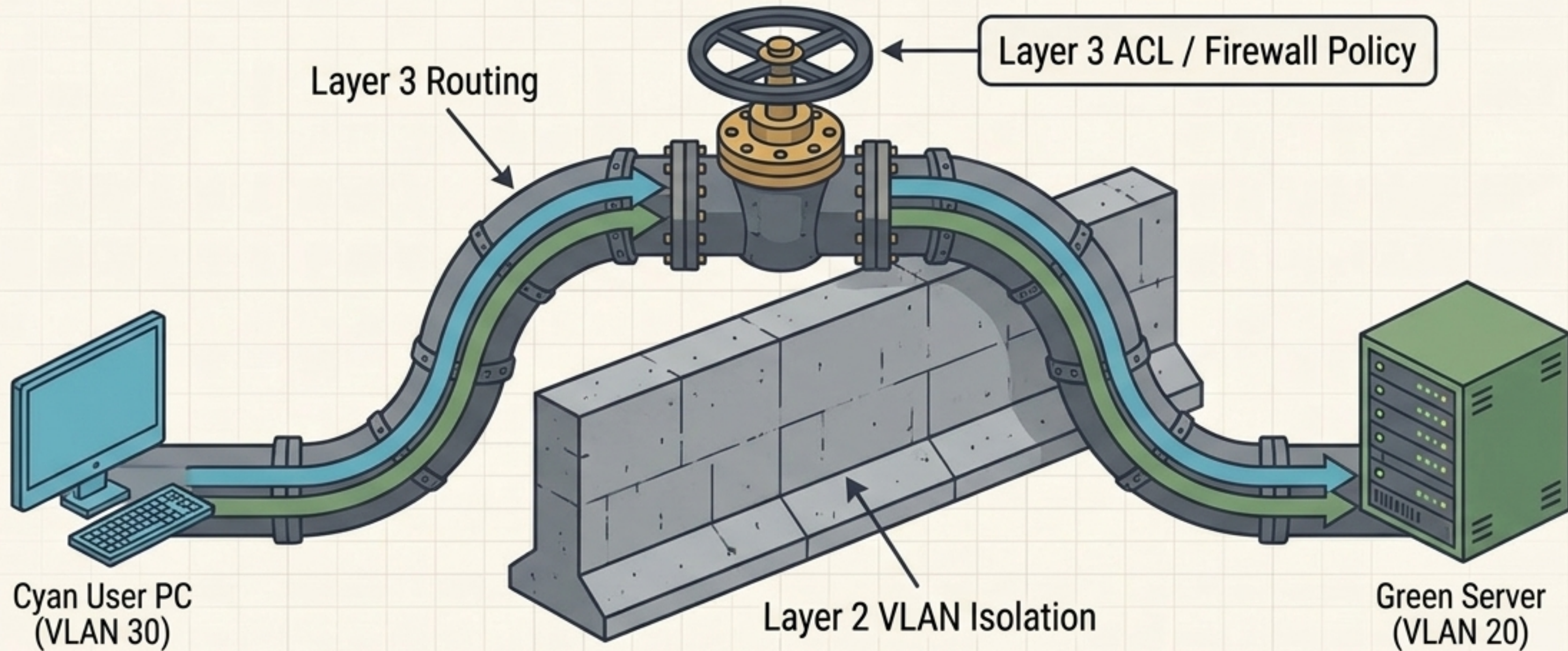
RoaS vs. L3 Switch

External Router 	Internal Routing 
Physical Trunk Link 	No Trunk Needed 
Subinterfaces 	SVIs 
Limited Scale 	Massive Enterprise Scale 



The Security Illusion

A critical design reality: VLANs define segmentation, not security. Once inter-VLAN routing is enabled, Layer 2 isolation is entirely bypassed.

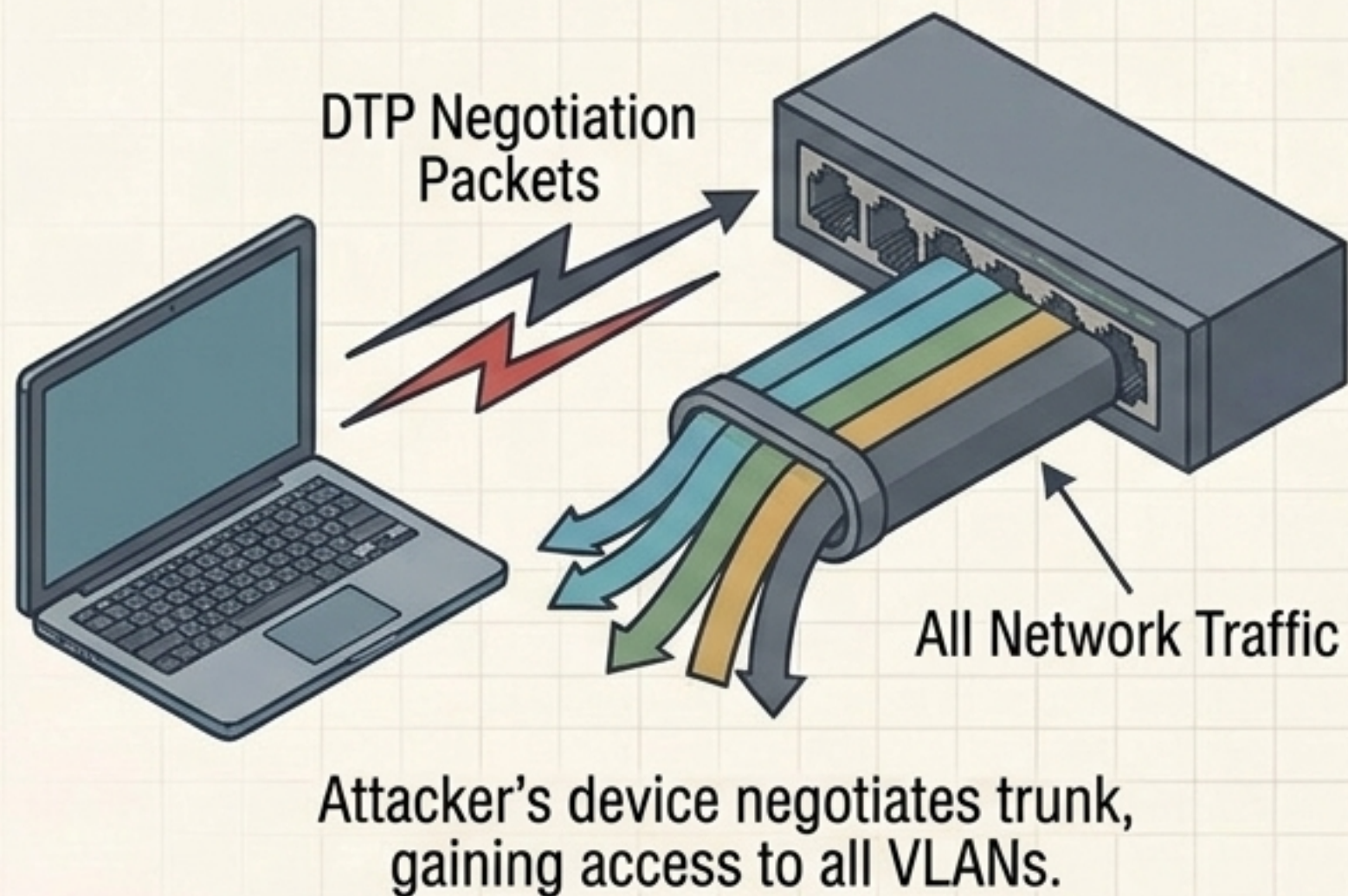


Takeaway: Security requires Layer 3 policies. Without ACLs or firewalls acting as the valve, routing nullifies your segmentation.

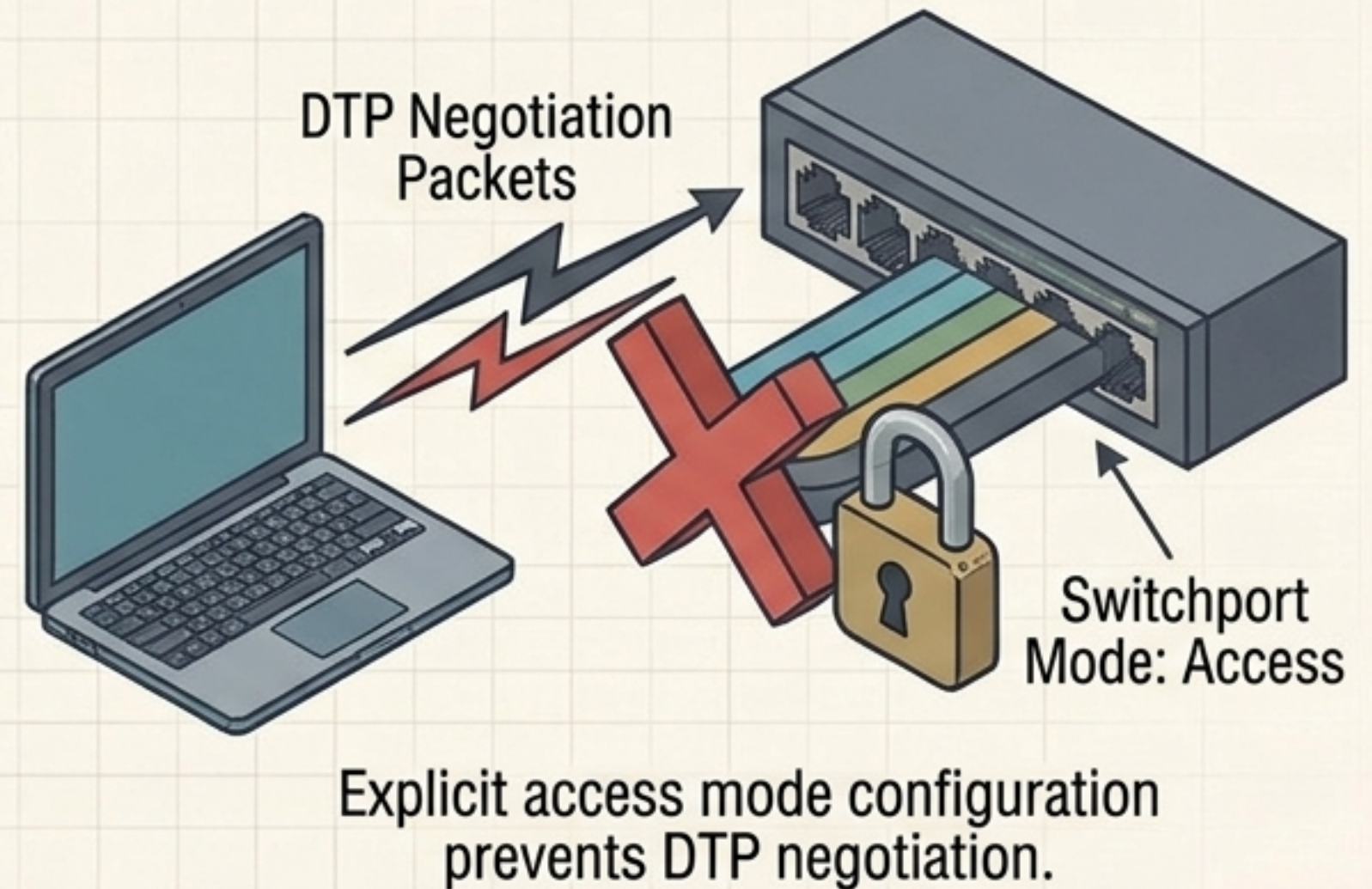
Vulnerability 1: Switch Spoofing

If an endpoint port is left to dynamically negotiate (Dynamic Trunking Protocol / DTP), an attacker can easily bypass access controls.

The Exploit



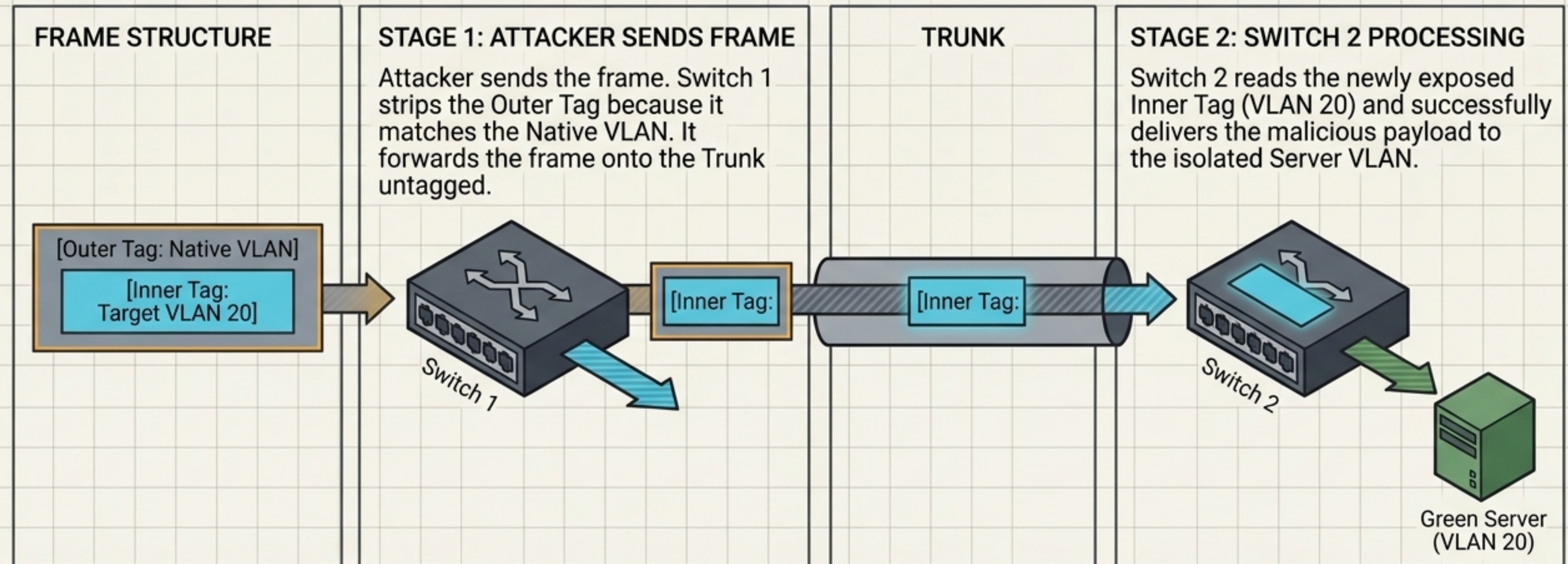
The Fix



Mitigation: Never trust default negotiation. Explicitly configure endpoint-facing ports as strict access ports (switchport mode access).

Vulnerability 2: Double-Tagging

An attacker crafts a malicious frame containing two 802.1Q tags to “hop” VLANs, exploiting how switches process Native VLAN traffic.



Anatomy of a Hardened Trunk

A secure trunk leaves absolutely nothing to default behavior. It is **explicitly configured, heavily restricted, and isolated**.

```
switchport mode trunk  
switchport trunk native vlan 999  
switchport trunk allowed vlan 10,20,30  
switchport nonegotiate
```

Explicit Definition: Disables dynamic negotiation (DTP).

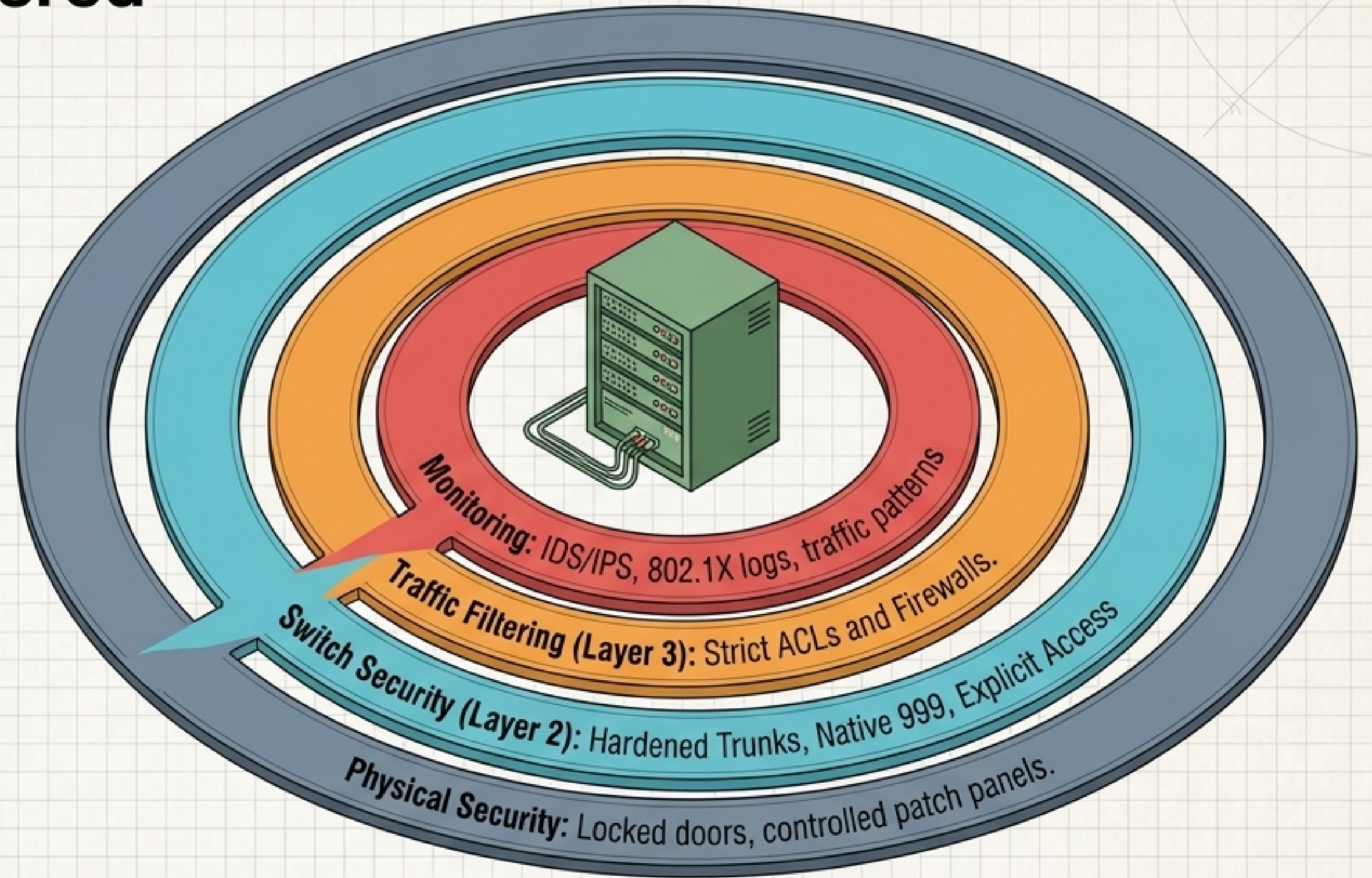
Isolated Native: Moves untagged traffic to a blackhole, preventing double-tagging exploits.

Least Privilege: Drops all unneeded VLANs at the port level.

Silence Protocols: Stops the interface from generating DTP frames entirely.

Synthesis: The Layered Defense Model

A resilient network architecture never relies on a single feature. True security is built by combining physical, logical, routing, and monitoring controls.



Final Takeaway:

Unnecessary trust creates unnecessary attack surface. Defend every layer.